SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 3.1: Acids and Bases — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Acids and Bases
Driving Question	DRIVING QUESTION: "What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Colour-Changing Drinks of the School Canteen — Launching the Phenomenon**	Students observed that common school-canteen liquids (lemon juice, uji, milk, soda, water, soap solution) produced strikingly different colours when drops of red cabbage indicator or universal indicator were added — ranging from bright red through purple to green and yellow. The same liquid always produced the same colour, and mixing certain liquids caused the colour to shift. Pedagogical note: Encourage students to record exact colours before any vocabulary is introduced; resist the urge to explain at this stage — raw observation data is the goal.	Students learned that liquids can be sorted into at least two broad groups based on the colour they produce with an indicator, and that this colour difference reflects an invisible chemical property of the liquid. They were introduced to the terms ACID and BASE as the labels chemists use for these two groups, and to INDICATOR as a substance that changes colour in the presence of an acid or a base. Pedagogical note: Post the two-column class anchor chart (Acids | Bases) on the wall — it will be updated every lesson.	The colour change is explained by the fact that indicator molecules are themselves weak acids or bases whose molecular and ionic forms absorb different wavelengths of light. When the indicator is placed in an acidic liquid, the equilibrium favours one coloured form; in a basic liquid, the other form dominates. At this stage students need only the surface-level explanation: acids and bases are real, invisible chemical properties that indicators make visible. The driving question is formally posted on the Driving Question Board. Pedagogical note: Do NOT resolve the driving question yet — use cold-calling (minimum 4 students) to surface pre-existing misconceptions such as 'colour means it is poisonous' or 'all acids burn you'.
Lesson 2 **What Are Acids and Bases? — Naming the Invisible**	Students observed the physical properties of a set of safe, dilute acid and base samples (lemon juice, dilute ethanoic acid/vinegar, dilute NaOH solution, bicarbonate of soda solution, tap water). They tested colour, smell (waft only), taste (lemon juice and uji only — never lab	Students learned the observable properties of acids (sour taste, turn blue litmus red, turn phenolphthalein colourless, turn methyl orange red, turn universal indicator red/orange/yellow) and bases/alkalis (bitter taste, slippery feel, turn red litmus blue, turn phenolphthalein pink/magenta, turn methyl	The different indicator colours are explained by the different ions each type of solution contains. Acids contain an excess of H^+ ions; bases/alkalis contain an excess of OH^- ions. These ions interact with the indicator molecule in opposite ways, producing opposite colour shifts. This

	chemicals), feel (dilute soap on fingertips), and litmus colour change. Results were recorded in a class data table. Pedagogical note: Emphasise SAFETY — only teacher-approved samples may be tasted or touched; lab chemicals are NEVER tasted.	orange yellow, turn universal indicator blue/green/purple). They also learned the distinction between a BASE (any substance that neutralises an acid) and an ALKALI (a base that dissolves in water). Pedagogical note: Use the mnemonic 'BRAT bases: Bitter, Red-litmus-blue, Alkali dissolves, Turn-phenolphthalein-pink' to help retention.	connects directly to the driving question: the 'invisible property' that makes lemon juice turn red and soap turn blue is the type and relative amount of ions present. Pedagogical note: Return to the anchor chart and add the property columns; ask students to update their DQB sticky notes.
Lesson 3 **What Is Inside an Acid or a Base? — The Arrhenius Definition**	Students observed a teacher-led demonstration of electrical conductivity: dilute HCl, dilute NaOH, ethanoic acid, and distilled water were each connected to a simple conductivity circuit (bulb/LED). Students recorded which solutions lit the bulb brightly, dimly, or not at all, and discussed what this implies about the presence of charged particles (ions) in solution. Pedagogical note: If a conductivity kit is unavailable, a PhET simulation or a video of the demonstration is an acceptable substitute for offline use on ARES.	Students learned the Arrhenius definitions: an ACID is a substance that produces H^+ ions (protons) when dissolved in water; a BASE is a substance that produces OH^- ions when dissolved in water. They practised writing dissociation equations for common acids ($HCl \rightarrow H^+ + Cl^-$; $H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$; $CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$) and bases ($NaOH \rightarrow Na^+ + OH^-$; $Ca(OH)_2 \rightarrow Ca^{2+} + 2OH^-$). Pedagogical note: Highlight the difference between the arrow ($\rightarrow$) for complete dissociation and the equilibrium arrows ($\rightleftharpoons$) for partial dissociation — this foreshadows Lesson 9.	The Arrhenius model explains WHY acids and bases behave differently: the H^+ ions from acids interact with indicators one way; the OH^- ions from bases interact the other way. This is the 'invisible property' the driving question refers to. Conductivity proves the ions are real — the brighter the bulb, the more ions in solution. Connect to Kenyan context: the hydrochloric acid in a student's stomach (gastric acid) is producing H^+ ions right now, enabling digestion of ugali and nyama choma. Pedagogical note: Cold-call at least 5 students to write one dissociation equation on the board; provide wait time of at least 10 seconds before accepting answers.
Lesson 4 **Differences Between Acids and Bases Using Commercial Indicators**	Students carried out a structured practical: they systematically tested a range of solutions (lemon juice, milk, bicarbonate solution, dilute HCl, dilute NaOH, Omo soap solution, black tea, Dettol diluted) with blue litmus, red litmus, phenolphthalein, and methyl orange. Results were recorded in a colour-coded class table. Pedagogical note: Pre-prepare dropper bottles labelled in Swahili and English to reinforce bilingual science literacy. Ensure each group has a results table with space for a written conclusion.	Students learned the specific, testable colour changes for each indicator: (1) Litmus — acids turn blue litmus RED, bases turn red litmus BLUE, neutral leaves both unchanged. (2) Phenolphthalein — colourless in acids and neutral, PINK/MAGENTA in bases. (3) Methyl orange — RED in acids, YELLOW in bases, ORANGE near neutral. They also reinforced that universal indicator gives a continuous colour spectrum (red → orange → yellow → green → blue → purple) corresponding to increasing basicity. Pedagogical note: The practical data from this lesson feeds directly into Lesson 8 (pH scale); keep class data tables posted for reference.	Each indicator has a different pH range over which it changes colour (its 'transition range'). Litmus changes around pH 7, methyl orange around pH 3–4, phenolphthalein around pH 8–10. Because different indicators change at different pH values, no single indicator tells the full story — universal indicator or a pH meter is needed for precise measurement. This explains why a farmer in Kericho testing tea-farm soil might use universal indicator rather than litmus alone. Pedagogical note: Link back to the driving question — the 'different colours' in the phenomenon are produced by these exact indicator–ion interactions.
Lesson 5 **Chemical Properties of Acids: Reactions with Metals and**	Students observed (live demo or video) two reactions: (1) zinc granules or magnesium ribbon added to dilute HCl — effervescence observed, burning splint test confirms	Students learned that: (1) Acids react with metals (zinc, magnesium, iron) to produce a SALT and HYDROGEN GAS. Balanced equation example: $Zn(s) + 2HCl(aq) \rightarrow$	In acid-metal reactions, the H^+ ions from the acid oxidise the metal, releasing hydrogen gas and forming metal ions that combine with the acid's anion to form a salt.

Neutralisation	hydrogen gas; (2) dilute HCl added dropwise to NaOH solution containing phenolphthalein — pink colour gradually disappears as neutralisation proceeds. Pedagogical note: The burning splint 'squeaky pop' test for H_2 is memorable and should be done live if possible; it becomes a long-term memory anchor for acid-metal reactions.	$ZnCl_2(aq) + H_2(g)$; $Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$. (2) Acids react with bases/alkalis (neutralisation) to produce a SALT and WATER ONLY. Balanced equation: $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$. Ionic equation: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$. Pedagogical note: Emphasise that the ionic equation for neutralisation is ALWAYS $H^+ + OH^- \rightarrow H_2O$ regardless of the specific acid or base used.	In neutralisation, H^+ and OH^- ions combine to form water, removing both from solution and producing a neutral (or near-neutral) result. Kenyan context: a farmer in the Rift Valley adding lime ($Ca(OH)_2$) to acidic soil is performing a large-scale neutralisation reaction — the same ionic chemistry, applied to food security. Pedagogical note: Ask students to predict (with wait time ≥10 s) what gas would be produced if magnesium were used instead of zinc — cold-call 3 students before confirming.
Lesson 6 **Acid Reactions with Carbonates and Hydrogen Carbonates — Completing the Acid Reaction Picture**	Students observed (practical or demo): (1) dilute HCl added to marble chips ($CaCO_3$) — vigorous effervescence, limewater turned milky confirming CO_2; (2) dilute HCl added to bicarbonate of soda ($NaHCO_3$) — effervescence, limewater test again confirms CO_2; (3) dilute HCl added to copper(II) oxide (CuO) powder with gentle heating — black powder dissolves to give blue/green copper(II) chloride solution. Pedagogical note: The limewater test for CO_2 ($Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$, milky precipitate) is a KICD-required practical skill; ensure every student records and can recall it.	Students learned that: (1) Acid + metal carbonate → salt + water + CO_2. Example: $2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$. (2) Acid + metal hydrogen carbonate → salt + water + CO_2. Example: $HCl(aq) + NaHCO_3(aq) \rightarrow NaCl(aq) + H_2O(l) + CO_2(g)$. (3) Acid + metal oxide → salt + water. Example: $H_2SO_4(aq) + CuO(s) \rightarrow CuSO_4(aq) + H_2O(l)$. Pedagogical note: Students often confuse carbonate and hydrogen carbonate reactions — both produce CO_2, both produce water, but stoichiometry differs; use side-by-side equation display.	The CO_2 produced in carbonate/hydrogen carbonate reactions comes from the carbonate ion (CO_3^{2-} or HCO_3^-) reacting with H^+ ions: $2H^+ + CO_3^{2-} \rightarrow H_2O + CO_2$. This is directly relevant to Kenyan daily life: the fizz in soda (carbonic acid H_2CO_3 releasing CO_2), antacid tablets ($NaHCO_3$ neutralising excess stomach acid), and baking mandazi ($NaHCO_3$ + acid in batter → CO_2 bubbles that make the dough rise). Pedagogical note: Connect to the driving question — 'why does it matter for our health?' — antacids are a direct answer; have students update the DQB.
Lesson 7 **Consolidation of Chemical Properties of Acids and Bases — Building the Reaction Summary Model**	Students reviewed all practical data collected in Lessons 5 and 6, completed a structured summary table (reaction type | reactants | products | test for gas/product | balanced equation | Kenyan example), and then participated in a 'reaction-type card sort' — matching reactant cards to product cards and equation cards. Pedagogical note: The card sort is a high-engagement retrieval practice activity; circulate and listen for misconceptions, especially confusion between 'salt' (ionic compound) and table salt (NaCl specifically).	Students consolidated the four chemical reaction types of acids: (1) Acid + Metal → Salt + H_2; (2) Acid + Base/Alkali → Salt + H_2O (neutralisation); (3) Acid + Metal Carbonate → Salt + H_2O + CO_2; (4) Acid + Metal Oxide → Salt + H_2O. They also practised naming salts from given acid-base pairs (HCl → chlorides; H_2SO_4 → sulfates; HNO_3 → nitrates; CH_3COOH → ethanoates). Pedagogical note: Salt naming is a frequent exam error; insist on the rule 'name the metal first, then the acid-derived anion' and give at least 6 varied examples.	All four reaction types share the common theme: H^+ ions from the acid react with another species (metal atoms, OH^- ions, CO_3^{2-}/HCO_3^- ions, or O^{2-} ions from metal oxides) and are consumed, producing a neutral or basic product. This unified model — 'acids react by donating H^+ ions' — is the Brønsted-Lowry view (introduced conceptually without formal labelling at this stage). Kenyan connection: the four reaction types map directly onto real Kenyan scenarios — stomach digestion, soil liming, mandazi baking, and industrial fertiliser production ($H_2SO_4 + NH_3$ → ammonium sulfate fertiliser used on Kenyan maize farms). Pedagogical note: Update the class cumulative model diagram on the wall.

Lesson 8 **Introduction to pH: Measuring the Strength of Acids and Bases in Kenyan Everyday Liquids**	Students used universal indicator solution (or pH paper) to measure the approximate pH of a curated set of Kenyan liquids: lemon juice, uji (fermented porridge), black tea, milk, tap water (Nairobi), bicarbonate solution, soap solution, and dilute bleach/Jik. Results were ranked on a 0–14 number line drawn on the board. Pedagogical note: If pH meters are available, use them alongside indicator to show agreement — this builds confidence in both methods. If neither is available, the ARES offline simulation of pH testing is an acceptable substitute.	Students learned that pH is a numerical scale from 0–14 measuring relative H^+ ion concentration in solution. pH < 7 = acidic; pH = 7 = neutral; pH > 7 = basic/alkaline. They learned that the scale is LOGARITHMIC — each unit represents a 10-fold change in H^+ concentration (a pH 3 solution has 10× more H^+ than a pH 4 solution). They matched the Kenyan liquids tested to approximate pH values and described health/agricultural significance of each. Pedagogical note: The logarithmic nature is conceptually challenging — use the analogy of decibels (loudness) or the Richter scale (earthquakes near Nairobi) to build intuition without requiring logarithm calculations at this stage.	pH explains the 'invisible property' at the heart of the driving question in a quantitative way. The colour changes observed in Lesson 1 were not random — they mapped onto specific pH values: lemon juice (pH ~2, red), black tea (pH ~5, orange-yellow), water (pH ~7, green), soap (pH ~9–10, blue). Health relevance: stomach acid pH ~1.5–2 enables protein digestion; blood must stay at pH 7.35–7.45 for survival. Agricultural relevance: maize grows best at pH 5.5–7.0; tea at pH 4.5–6.0 in Kericho; soil pH outside these ranges reduces crop yields. Environmental relevance: acid rain (pH ~4–5) acidifies Lake Victoria's tributaries, threatening tilapia. Pedagogical note: Have students answer the driving question partially in writing — collect as a formative assessment.
Lesson 9 **Strong vs. Weak Acids and Bases: Why the Same pH Step Can Make or Break a Tea Farm**	Students observed a teacher demonstration comparing equal concentrations (e.g., 0.1 mol/dm³) of HCl and ethanoic acid (vinegar) using: (1) conductivity test — HCl lights the bulb much more brightly; (2) pH paper — HCl gives pH ~1, ethanoic acid gives pH ~3; (3) rate of reaction with zinc granules — HCl reacts much faster. Pedagogical note: This trio of observations powerfully demonstrates that same concentration ≠ same strength; students often hold the misconception that 'more concentrated = stronger acid' — this lesson directly confronts it.	Students learned the distinction between STRONG and WEAK acids/bases. Strong acids (HCl, H_2SO_4, HNO_3) and strong bases (NaOH, KOH, $Ca(OH)_2$) dissociate COMPLETELY in aqueous solution, producing the maximum possible H^+ or OH^- ion concentration for a given concentration. Weak acids (ethanoic acid CH_3COOH, carbonic acid H_2CO_3, citric acid in lemon juice) and weak bases (NH_3 solution, $NaHCO_3$) dissociate only PARTIALLY, represented by equilibrium arrows (⇌). They used the terms DEGREE OF DISSOCIATION and understood that strength (strong/weak) and concentration (mol/dm³) are independent variables. Pedagogical note: Insist students practise saying and writing both terms together — 'concentrated weak acid' and 'dilute strong acid' are both valid and mean very different things.	The tea farms of Kericho and Limuru require soil pH of 4.5–6.0. A weak acid like tannic acid (naturally present in tea leaves) keeps the soil in this range because it only partially dissociates — it acts as a pH buffer. If a strong acid like H_2SO_4 from acid rain entered the soil at the same concentration, it would fully dissociate and crash the pH, destroying the crop. This is why the distinction between strong and weak acids matters for Kenyan agriculture. Pedagogical note: Cold-call at least 4 students to explain in their own words why HCl and CH_3COOH at equal concentration have different pH values — accept and refine student language before giving the formal explanation.
Lesson 10 **Uses of Acids and Bases + Final Model Build & Explanation**	Students reviewed all experimental data, indicator results, pH measurements, reaction products, and model diagrams collected across all 10 lessons. They then completed a final 'Uses Gallery Walk':	Students learned the major uses of acids and bases in Kenyan daily life: HCl (pH ~1–2) in gastric juice digests proteins in ugali, beans, and nyama choma; carbonic acid (H_2CO_3, pH ~4–5) gives sodas their fizz	The driving question is now fully answered: the invisible properties are the type and concentration of H^+ and OH^- ions (measurable as pH), which are produced when acids (Arrhenius: produce H^+) or

	posters around the room linked specific acids and bases (with pH values and reaction types) to Kenyan health, agricultural, industrial, and environmental contexts. Pedagogical note: The Gallery Walk should include at least 6 stations — stomach acid/digestion, soil liming, food preservation (citric acid in mango/passion fruit), industrial manufacture (H_2SO_4 in battery acid for matatu fleets), cleaning products (NaOH in soap), and water treatment (chlorine/Cl_2 acidifies water for Nairobi Water Works).	and acts as a blood pH buffer; citric acid preserves passion fruit and mango juice; H_2SO_4 is used in car/matatu batteries and to manufacture fertilisers; NaOH is used to make soap and process cassava; $Ca(OH)_2$ (lime, pH ~12) is used by farmers to neutralise acidic soil in the Rift Valley and by Nairobi City Water to treat drinking water; $NaHCO_3$ (pH ~8.3) is used as baking powder in mandazi and as an antacid. Pedagogical note: Require students to state BOTH the acid/base name AND its pH AND the relevant reaction type for each use — surface-level recall is insufficient.	bases (Arrhenius: produce OH^-) dissolve in water. These ions interact with indicator molecules to produce colour changes. The pH value matters for health (gastric acid, blood pH), agriculture (soil pH for maize, beans, and tea), and environment (acid rain effects on Lake Victoria and Kenyan highland rivers). In the final model, students annotate a cumulative diagram showing: indicator colour ↔ pH value ↔ ion concentration ↔ acid/base classification ↔ reaction type ↔ real-world Kenyan use. Pedagogical note: Collect final written explanations as a summative formative assessment; use the driving question as the essay prompt and assess against the three-part criterion: health, farming, and environment.